Insurance Cost Analysis

```r
insurance <- read.csv("data/medical_insurance.csv")
names(insurance)
```

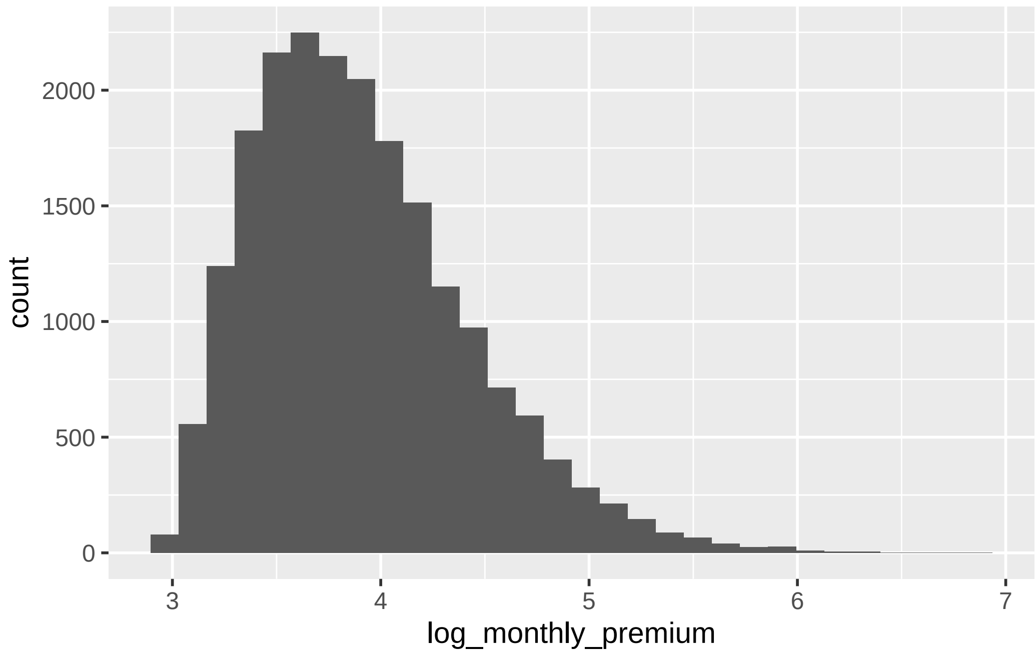
```
 [1] "person_id"                   "age"
 [3] "sex"                         "region"
 [5] "urban_rural"                 "income"
 [7] "education"                   "marital_status"
 [9] "employment_status"           "household_size"
[11] "dependents"                  "bmi"
[13] "smoker"                      "alcohol_freq"
[15] "visits_last_year"            "hospitalizations_last_3yrs"
[17] "days_hospitalized_last_3yrs" "medication_count"
[19] "systolic_bp"                 "diastolic_bp"
[21] "ldl"                         "hba1c"
[23] "plan_type"                   "network_tier"
[25] "deductible"                  "copay"
[27] "policy_term_years"           "policy_changes_last_2yrs"
[29] "provider_quality"            "risk_score"
[31] "annual_medical_cost"         "annual_premium"
[33] "monthly_premium"             "claims_count"
[35] "avg_claim_amount"            "total_claims_paid"
[37] "chronic_count"               "hypertension"
[39] "diabetes"                    "asthma"
[41] "copd"                        "cardiovascular_disease"
[43] "cancer_history"              "kidney_disease"
[45] "liver_disease"               "arthritis"
[47] "mental_health"               "proc_imaging_count"
[49] "proc_surgery_count"          "proc_physio_count"
[51] "proc_consult_count"          "proc_lab_count"
[53] "is_high_risk"                "had_major_procedure"
```

What proportion of insurance cost variance is attributable to modifiable lifestyle choices versus baseline demographic and socioeconomic risk factors?

Modifiable lifestyle choices

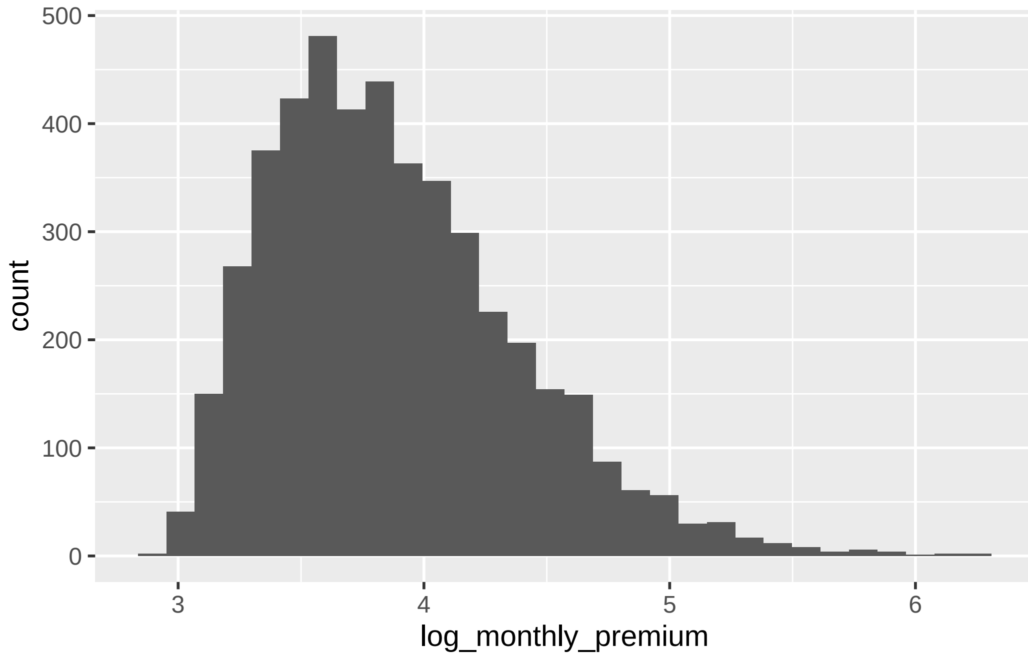
```
insurance <- mutate(insurance, health = if_else(hypertension == 1, "hypertension", if_else(as

insurance <- mutate(insurance, log_monthly_premium = log(monthly_premium))

#Created new column named health displaying medical conditions of individuals, filtered for

insurancehpt <- insurance |> filter(health == "hypertension")
ggplot(data = insurancehpt, mapping = aes(x = log_monthly_premium)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



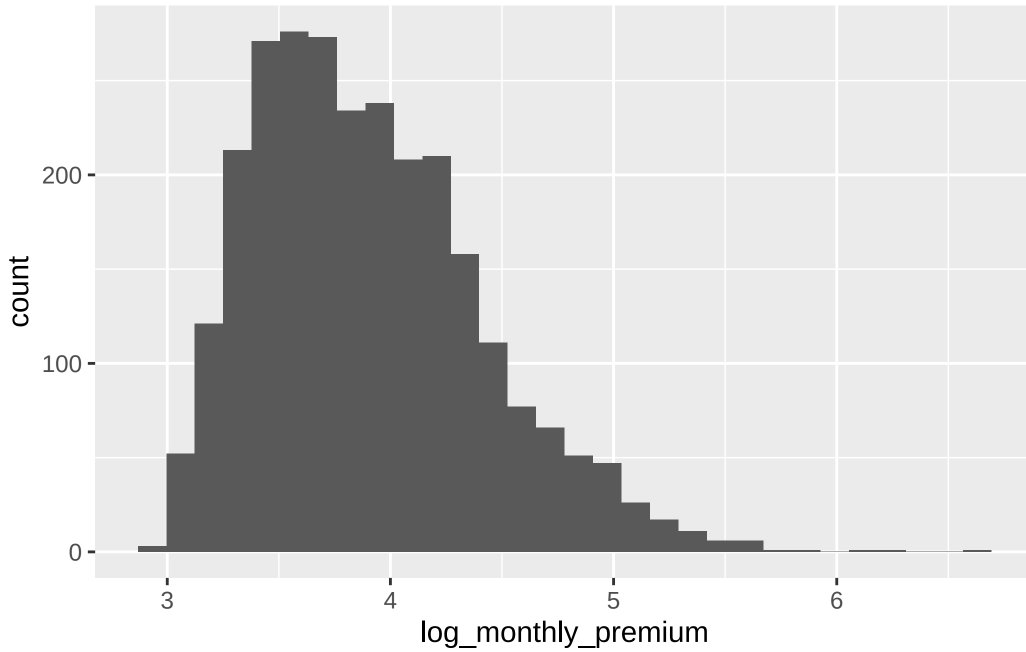

```
insuranceast <- insurance |> filter(health == "asthma")
ggplot(data = insuranceast, mapping = aes(x = log_monthly_premium)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



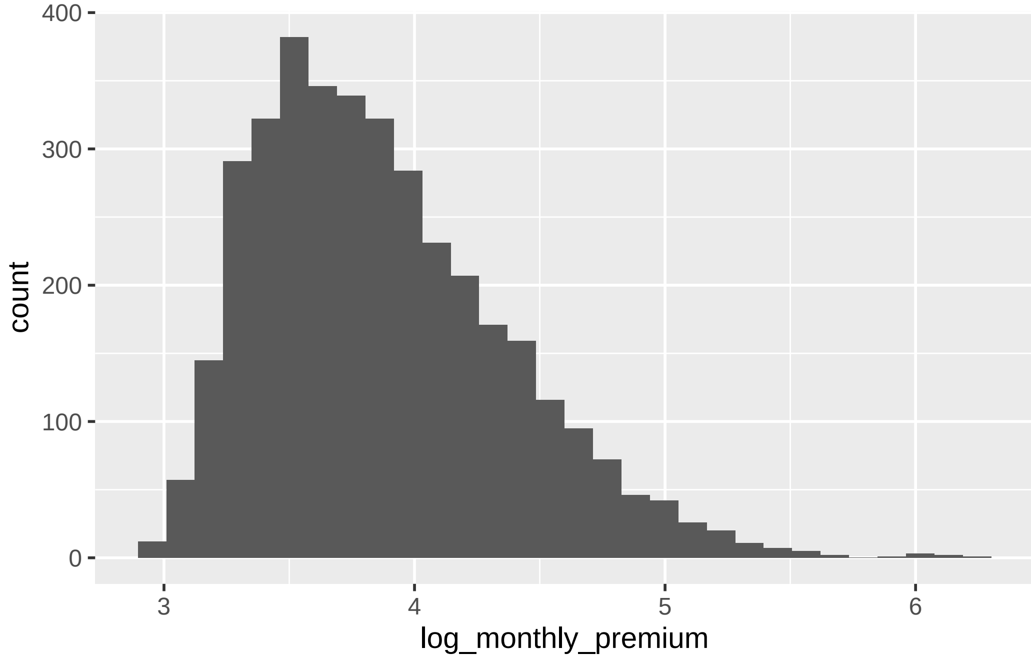

```
insurancecopd <- insurance |> filter(health == "copd")
ggplot(data = insurancecopd, mapping = aes(x = log_monthly_premium)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



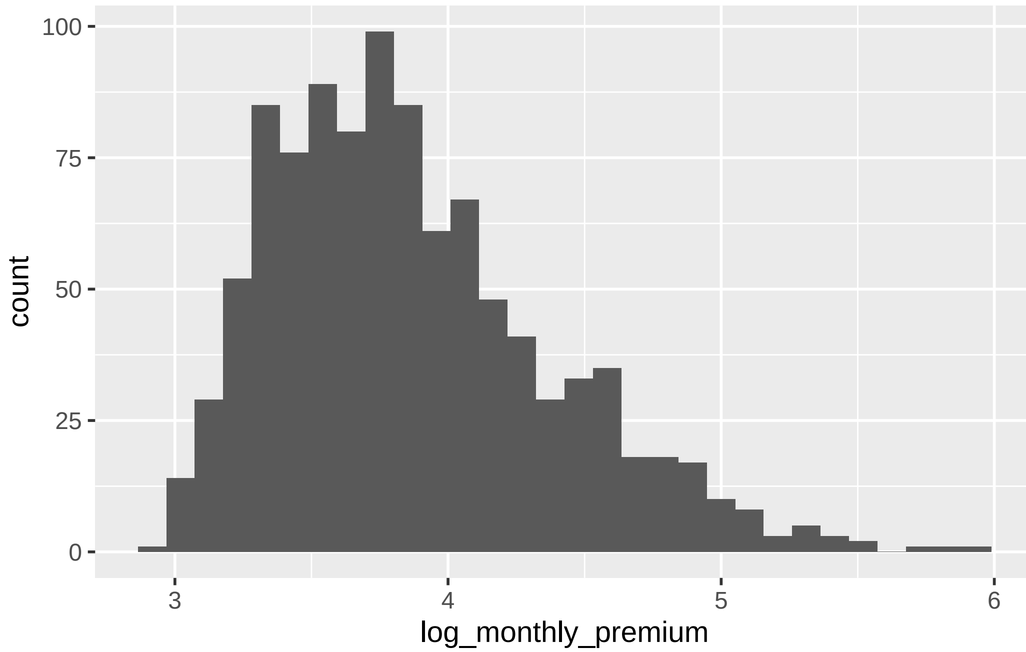

```
insurancecard <- insurance |> filter(health == "cardiovascular disease")
ggplot(data = insurancecard, mapping = aes(x = log_monthly_premium)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



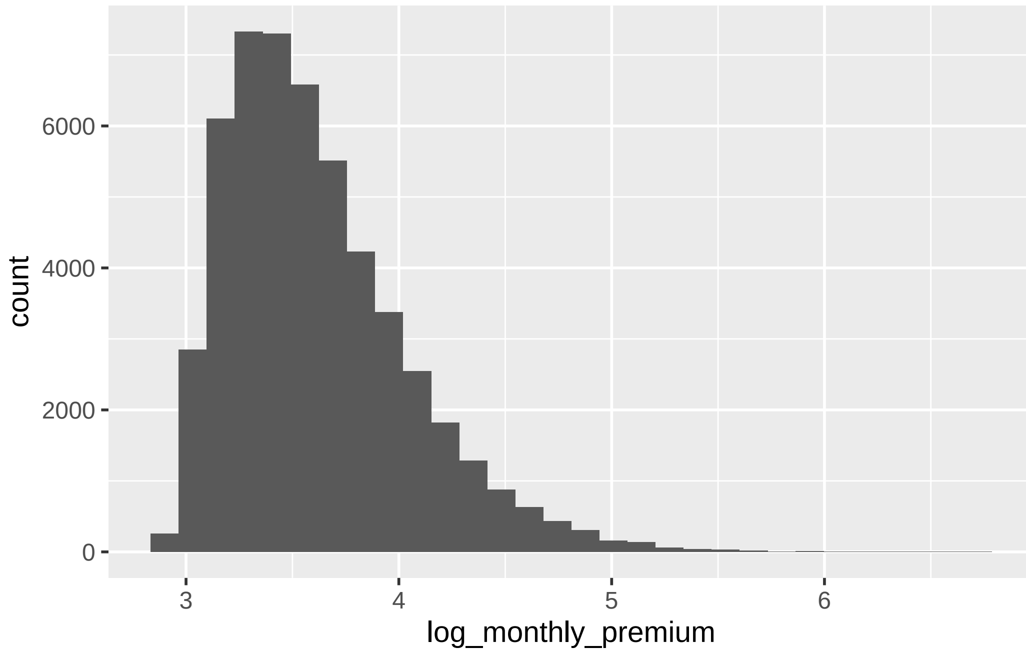

```
insurancekidn <- insurance |> filter(health == "kidney disease")
ggplot(data = insurancekidn, mapping = aes(x = log_monthly_premium)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
insuranceother <- insurance |> filter(health == "other")
ggplot(data = insuranceother, mapping = aes(x = log_monthly_premium)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```r
summary(aov(log_monthly_premium ~ health, data = insurance))
```

```
               Df Sum Sq Mean Sq F value Pr(>F)
health          8   1838  229.69    1061 <2e-16 ***
Residuals   99991  21649    0.22
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```